UNIVERSITY OF BRISTOL
MSC BIOINFORMATICS

Evaluating Evapotranspiration Generalisation Under a Changing Climate using Machine Learning

Review and Plan by
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ABSTRACT (The abstract has a maximum of 200 words)

- Summary of background on ET/ML methods and FLUXNET
- Overall aim of the project
- Proposed approach for the project

Key words: Machine Learning, FLUXNET, Evapotranspiration.
Word count: 3000

Contents

Literature Review (~ 1200 words)

Introduction (~ 500 words)

Evapotranspiration

- Overview of evapotranspiration (ET):

$$ET = Transpiration + E_{Canopy} + E_{Soil} \tag{1}$$

- Drivers of ET e.g Leaf Area index (LAi), Vapour pressure deficit (VPD), precipitation, air temperature etc.
- Importance and applications of ET predictability

Leaf Area Index

- What is LAi and how is it measured?
- How is this related to ET?

FLUXNET

- Overview of eddy covariance and how flux towers collect data
- Brief look at FLUXNET and how sites are grouped (region and vegetation types)

The Changing Climate

- Overview of climate change and effects on carbon dioxide levels
- How these could link to factors such as changing ET

Contemporary ET Prediction Methods (~ 450 words)

Non-ML methods

- Quick look at some of these methods. List below from Xu et al. [2018]:
- Predictive equations which link ET observations (flux towers) to vegetation indices (e.g LAi), land surface temp. observations, and meteorological parameters
- "Geostatistical methods" based on kriging theoretical framework
- Using semi-theoretical or running theoretical methods

ML methods

- Brief overview of ML methods
- Performances of different models in the literature

Knowledge Gaps and Project Scope (~ 250 words)

Knowledge Gaps and Unresolved Questions

- Findings from literature
- Address gaps in the literature
- Suggest areas for future efforts (which are not this project) - ?

Project Scope

- Suggest the area/question this project is going to address
- Include why this is relevant currently/addressable now
- "Do predictive ET ML models trained on historic FLUXNET and satellite data have a loss of performance under new climate conditions?"

Research Project Plan (~ 1500 words)

Overall Project Aim (~ 200 words)

- Overview of how this project will answer the question/area proposed at end of introduction.

Methods Overview (~ 1300 words)

Data Acquisition and Selection

- How the data will be accessed and stored (remote sensing and FLUXNET).
- Time period (requires long term sites) and geographical location of data selected for modelling.
- Why these were selected and appropriate? (in terms of location/times, not specific structure).

Feature Engineering

- Selection of data features for use in training and why.
- How these were removed or adapted
- Why are these features appropriate?

ML Model Setup and Training

- How the data will be accessed and stored (remote sensing and FLUXNET).
- Explanation of model(s) selection (type of ML method)
- e.g If including LAi vs. not with other parameters kept identical across 2 models.
- Overview of model parameters (- might be too much detail?) and how each are trained.
- Why are these model choices/parameters appropriate for this task?
- Target features: FLUXNET latent heat flux (ET approximation)
- Predictor features: Meteorological data (from FLUXNET and others like LAi from satellite data e.g ECoSTRESS)

ML Model Evaluation

- Testing methods on unseen/held out historical (same as training e.g 5-20 years old) validation sets
- Comparison to other predictive models e.g in FLUXCOM
- Overall evaluation of model predictability on historical data

ML Model Predictions

- Predictions made on unseen and more recent data (e.g last 5 years)
- Overall evaluation of model predictability on recent data

Gantt Chart

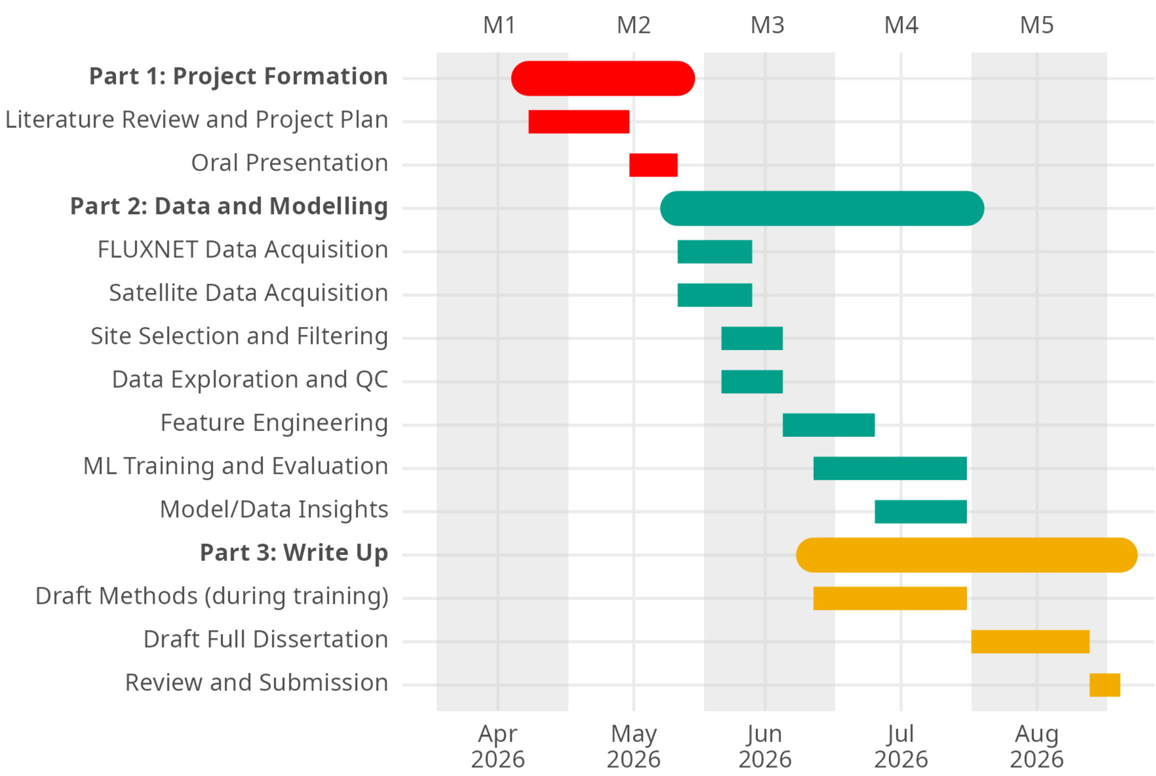

Figure 1: A proposed Gantt Chart for the duration of this project.

Risk Register (~ 300 words)

- 4-5 risks at about 60 words each
- Data access issues, overfitting, data gaps etc.

References

Xu, T., Guo, Z., Liu, S., He, X., Meng, Y., Xu, Z., Xia, Y., Xiao, J., Zhang, Y., Ma, Y., et al. (2018). Evaluating different machine learning methods for upscaling evapotranspiration from flux towers to the regional scale. *Journal of Geophysical Research: Atmospheres*, 123(16):8674–8690.